CLAIMS

What is claimed is:

1. A system for high-radix computing, comprising:

    - a processing unit integrated on a semiconductor die;
    - wherein the processing unit comprises a plurality of phototransistors, each phototransistor configured to receive an optical signal and to distinguish between a plurality of discrete wavelengths of light; and
    - wherein each of said discrete wavelengths corresponds to a unique logical state in a number system with a base greater than two.

2. The system of claim 1, wherein the processing unit further comprises an arithmetic logic unit (ALU) and one or more registers constructed from said phototransistors, configured to perform computations directly on data represented in said number system.

3. The system of claim 1, further comprising an optical interface integrated on the same semiconductor die, configured to couple the processing unit to a parallel optical bus.

4. The system of claim 3, wherein the system is configured to receive a high-radix data word from the optical bus and load it into a register in a single clock cycle.

5. The system of claim 1, wherein the number system is a base-4 system or higher.

6. A method for processing data, comprising the steps of:

    - receiving an optical signal comprising one of a plurality of discrete wavelengths at a phototransistor-based logic element;

- identifying, at said logic element, which of the plurality of discrete wavelengths was received;
- assigning a unique logical state to the received optical signal based on the identified wavelength, wherein said logical state is part of a number system with a base greater than two; and
- performing a computation using said logical state as an operand.

7. The method of claim 6, wherein the computation is performed by a high-radix arithmetic logic unit (ALU) integrated on the same die as the logic element.

8. The method of claim 6, wherein the optical signal is received directly from a massively parallel optical bus.